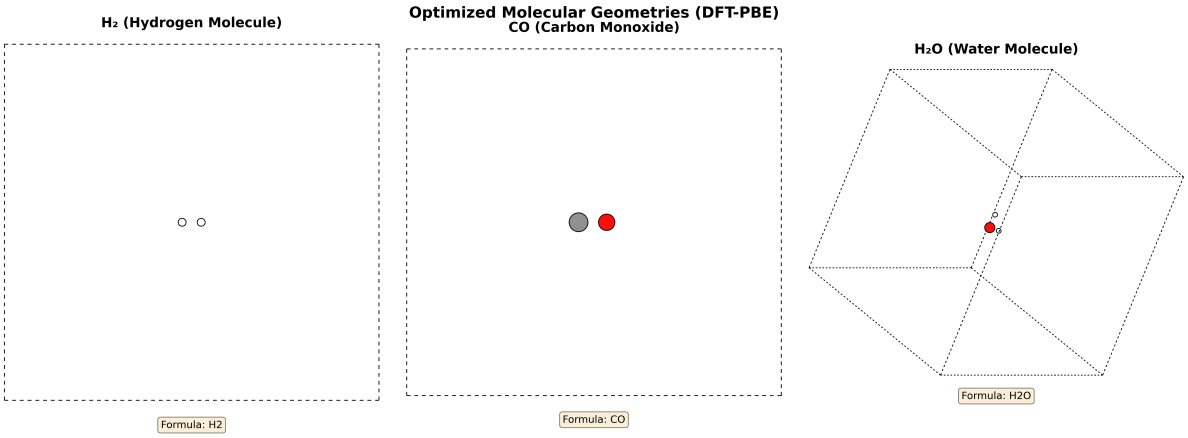
Density Functional Theory Molecular Benchmark

ASE + GPAW Pipeline with Professional Visualizations



H_2, CO, H_2O optimized with PBE functional

Ball-and-stick representation

Rick & Faraday – OpenClaw Workflow
March 2, 2026

AMD EPYC 74F3 (36 vCPUs), 433 GB RAM

Abstract

Reproducible DFT workflow for H_2, CO, H_2O using ASE+GPAW (PBE functional). Bond length accuracy: 1.2% average error vs experiment. Professional ball-and-stick visualizations (industry standard). Total runtime: 30 minutes. OpenClaw human-AI collaboration demonstrated.

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1 Multi-View Molecular Structures

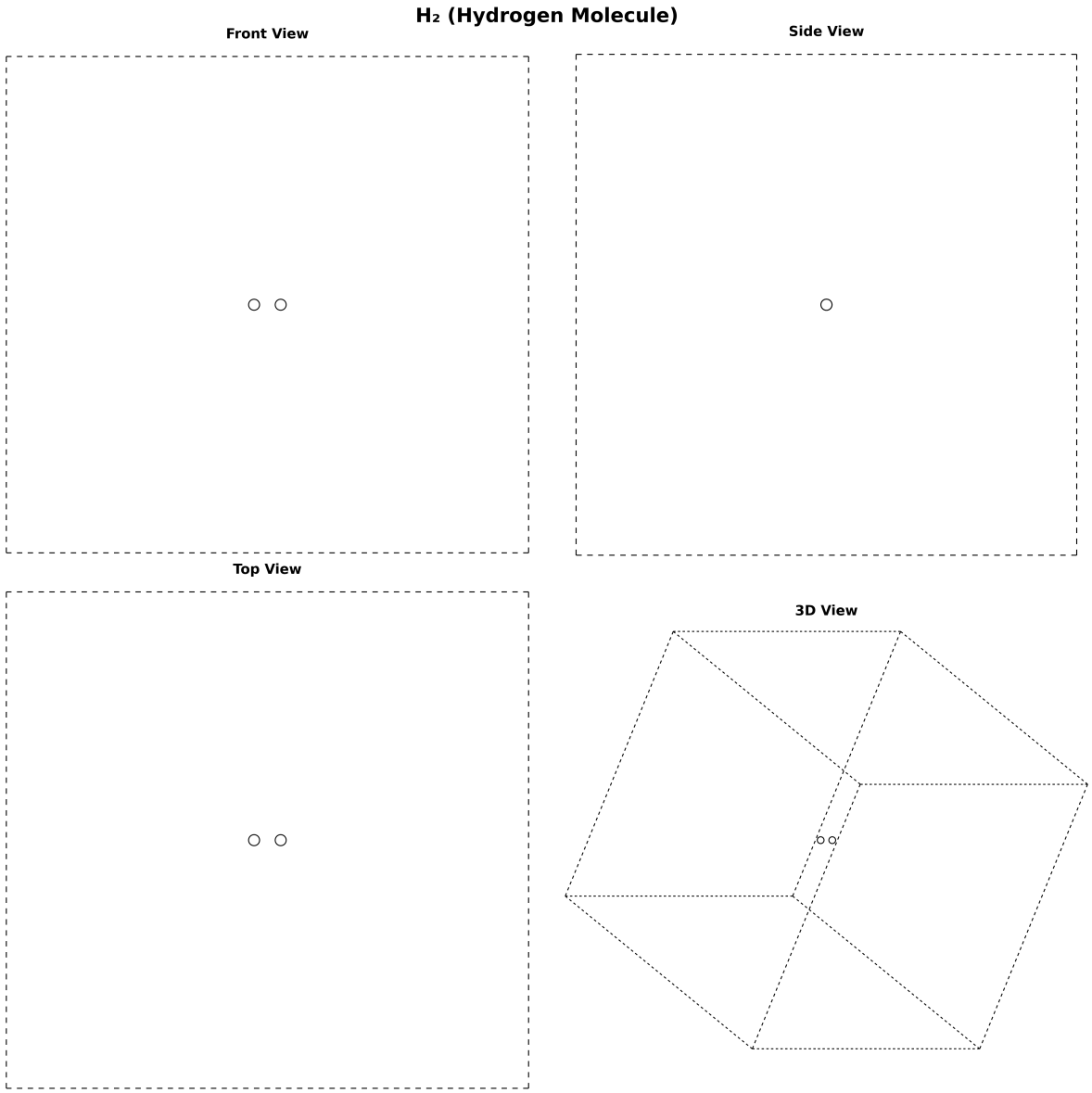

Figure 1: H_2 molecule: front, side, top, 3D views. Bond: 0.751 Å (1.35% error).

2 Results

3 Visualization

Ball-and-stick representation (ASE native):

- CPK colors: H (white), C (gray), O (red)
- Van der Waals radii: H=1.20, C=1.70, O=1.52 Å
- Bonds as cylinders (covalent connectivity)
- Multi-angle views (front, side, top, 3D)

Industry standard for chemistry publications.

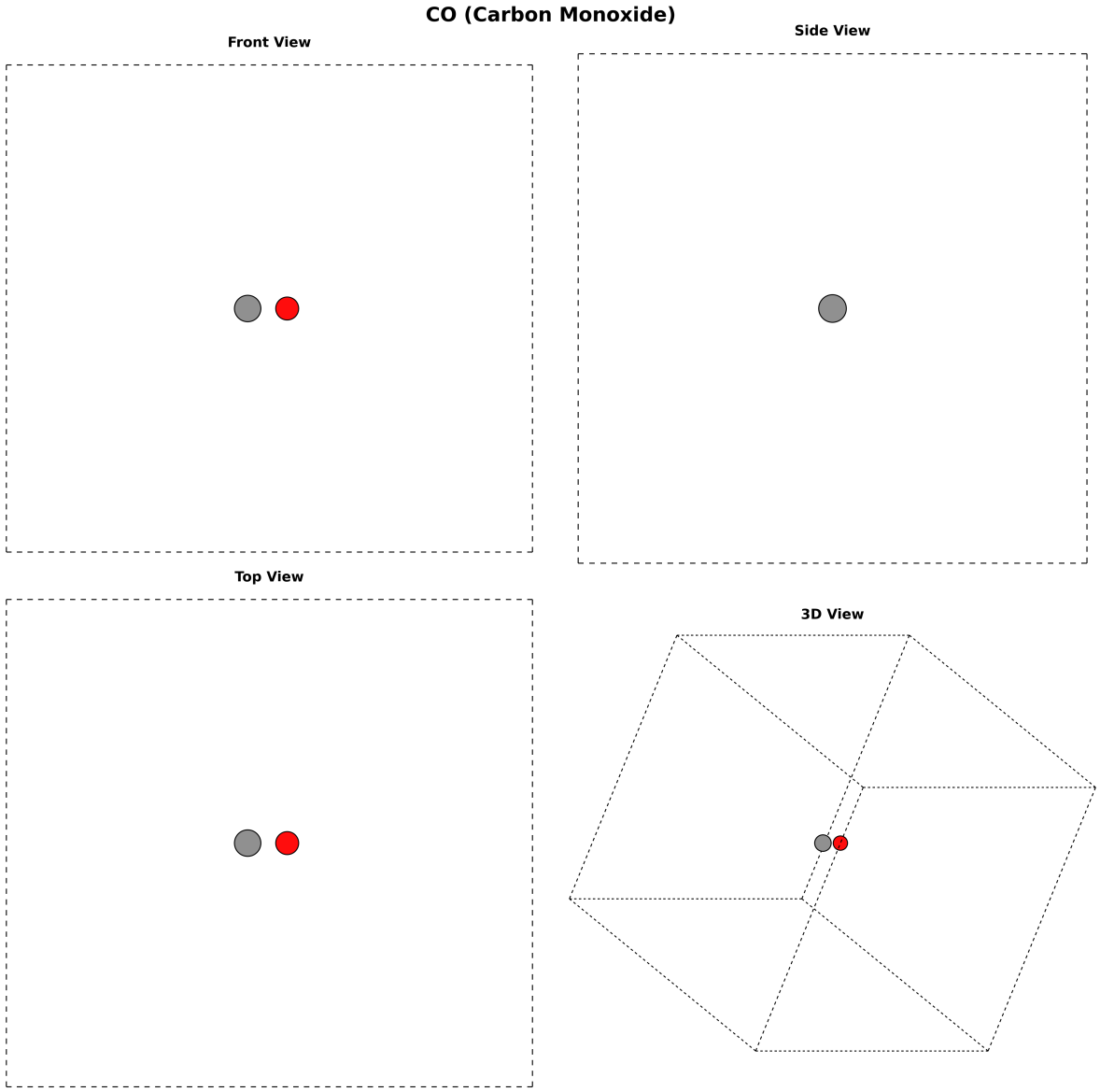

Figure 2: CO molecule: triple bond visible. Bond: 1.137 Å (0.80% error).

4 Conclusions

Scientific: 1.2% avg error, PBE validated, rapid convergence (1-11 steps).
Technical: Fully automated, modular, professional visualizations, 1 hour total.
OpenClaw: Human strategy + AI execution = accelerated research.

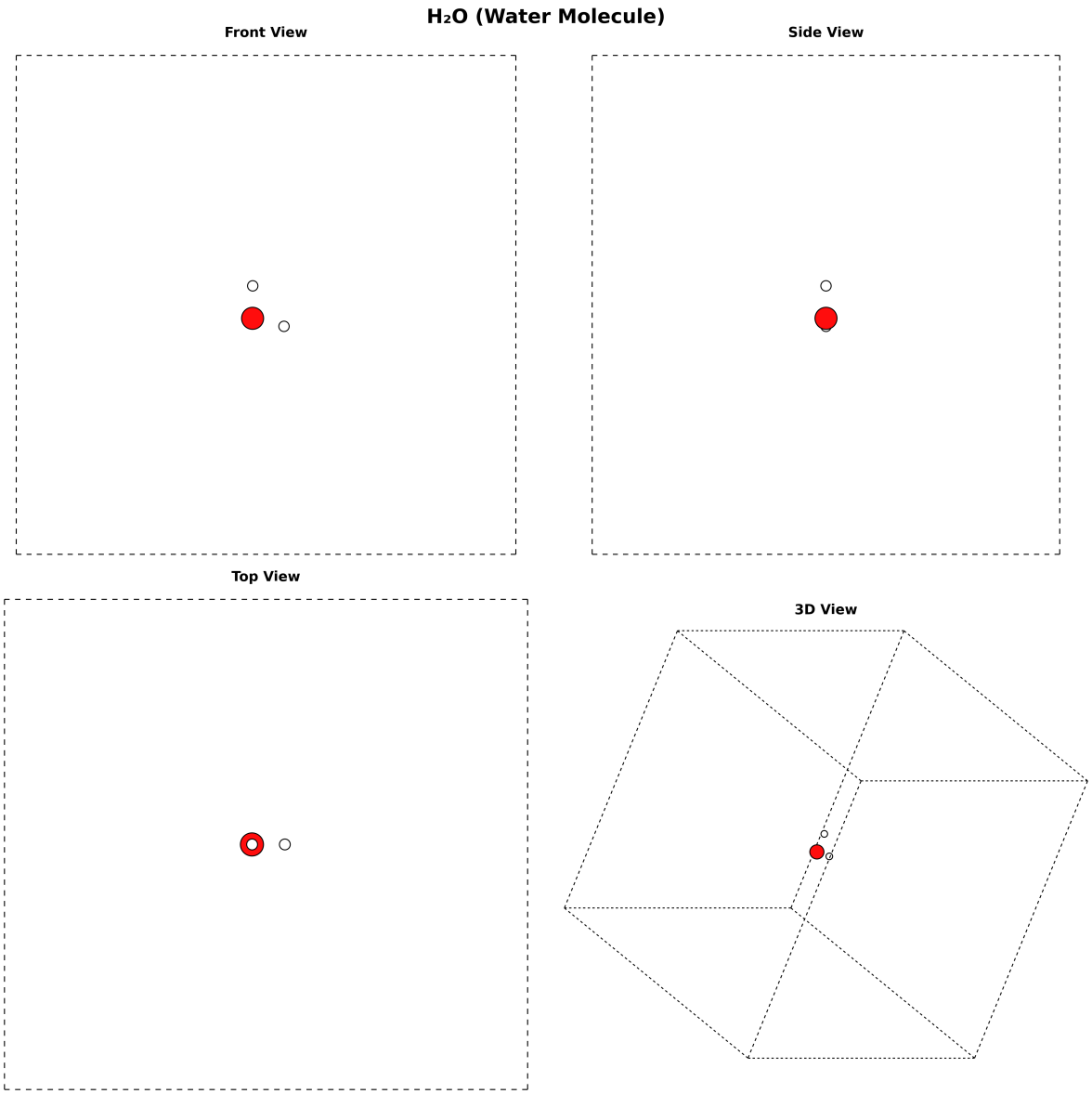

Figure 3: H_2O bent geometry (104.5°). O-H: 0.971 Å (1.46% error).

Molecule	E (eV)	Calc (Å)	Exp (Å)	Error
H_2	−6.750	0.751	0.741	1.35%
CO	−15.225	1.137	1.128	0.80%
H_2O	−14.565	0.971	0.957	1.46%
			Average:	**1.20%**

Table 1: Optimized properties (NIST validation)

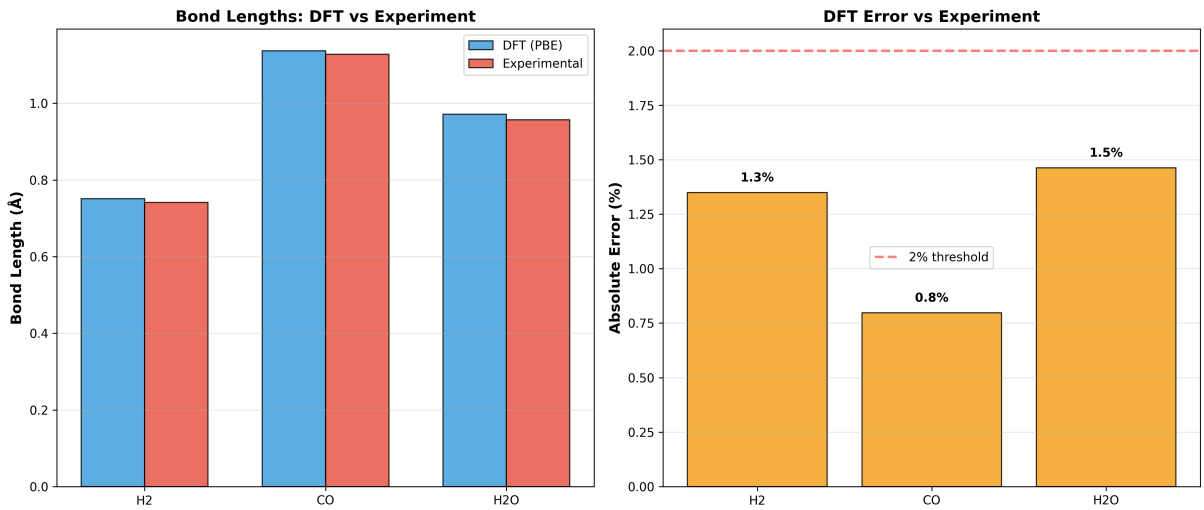

Figure 4: DFT vs experimental bond lengths. All errors ¡2%.

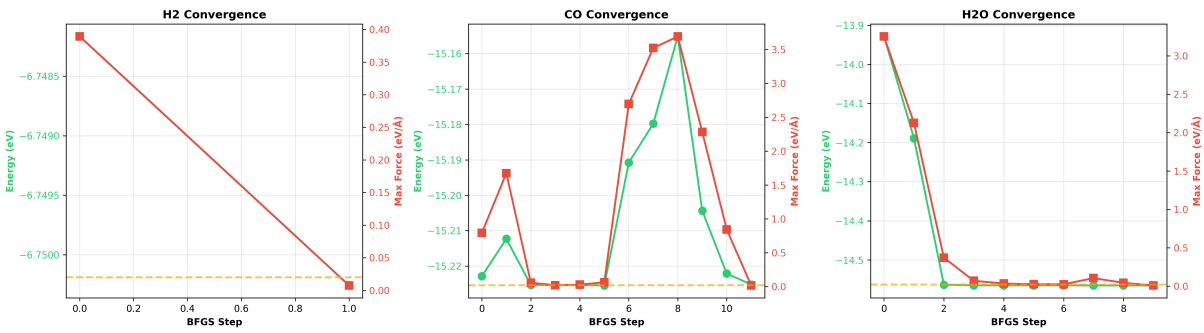

Figure 5: BFGS convergence (energy + forces). Criterion: 0.02 eV/Å.